

What You Always Wanted to Know About Evolution Strategies, But Never Dared to Ask

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Instructor

Hans-Georg Beyer is best known for his theoretical analyses and the design of Evolution Strategies based on the stochastic dynamical systems approach. Dr. Beyer received the Diploma degree in Theoretical Electrical Engineering from the Ilmenau Technical University, Germany, in 1982 and the Ph.D. in physics from Bauhaus-University Weimar, Weimar, Germany, in 1989, and the Habilitation degree in computer science from the University of Dortmund, Dortmund, Germany, in 1997. He was an R&D Engineer with the Reliability Physics Department, VEB Gleichrichterwerk, Stahnsdorf, Germany, from 1982 to 1984. From 1984 to 1989, he was a Research and Teaching Assistant and later on Post-Doctoral Research Fellow with the Physics Department and the Computer Science Department, Bauhaus-University Weimar. From 1990 to 1992, he was a Senior Researcher with the Electromagnetic Fields Theory Group, Darmstadt University of Technology, Darmstadt, Germany. From 1993 to 2004, he was with the Computer Science Department, University of Dortmund. In 1997, he became a DFG (German Research Foundation) Heisenberg Fellow. He was leader of a working group and a Professor of Computer Science from 2003 to 2004. Since 2004 he has been professor with the Vorarlberg University of Applied Sciences, Dornbirn, Austria. He authored the book "The Theory of Evolution Strategies" (Heidelberg: Springer-Verlag, 2001) and authored/coauthored over 200 papers. Dr. Beyer was the Editor-in-Chief of the MIT Press Journal "Evolutionary Computation" from 2010 to 2016. He was an Associate Editor for the IEEE "Transactions on Evolutionary Computation" from 1997 to 2021 and is a member of the advisory board of the Elsevier journal "Swarm and Evolutionary Computation" and MIT Press "Evolutionary Computation."



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Evolution Strategies Revisited

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Agenda

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- 3 A Short Recap of Evolution Strategies
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Evolution Strategies Revisited

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The Flowchart of Darwinian Evolution

The Flowchart of DARWINian Evolution

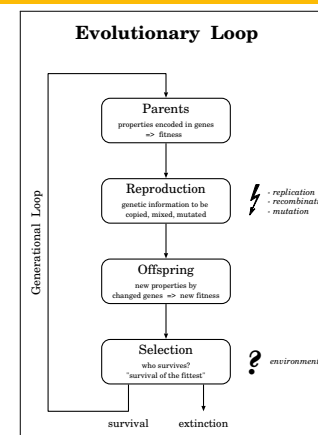


Figure 1: The “algorithm” of DARWINian evolution.

- driving evolution principle:
VARIATION-SELECTION-PRINCIPLE
- rarely questioned conclusion:
Darwinian Evolution = Optimization
- Evolutionary Algorithms (EA)
- different EA flavors are due to different variation and selection methods:
- **classical Evolution Strategies (ES)** typically use **truncation selection** in contrast to, e.g., classical Genetic Algorithms (GA)
- variation in ES, especially **mutation**, is **by sampling from a distribution** in contrast to, e.g., classical Differential Evolution (DE) and GA

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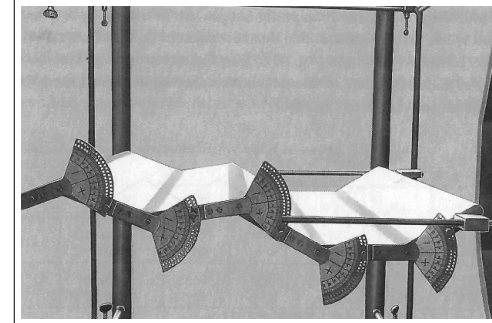
Evolution Strategies Revisited

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How it All Began - History of ES

- first ES experiments were done on real physical objects (!)
also referred to as *hardware in the loop optimization*
- date back to summer 1964:
“**Experimentum Crucis:**” Evolution of a kinked plate
- in 1965: evolution of a pipe bend
- in 1968: evolution of a two-phase nozzle
- first computer experiments were performed in 1965
- computer experiments were done in $\mathbb{R}^N$, but also in $\mathbb{B}^N$,
☞ mimicry experiment [Slide 12](#)
- ➔ optimization of material objects has built the philosophical basis of ES

The Experimentum Crucis (Summer, 1964¹)



(reproduced from I. Rechenberg [1], *Evolutionsstrategie: Optimierung technischer Systeme nach Prinzipien der biologischen Evolution*. Frommann-Holzboog, Stuttgart, 1973, p. 25.)

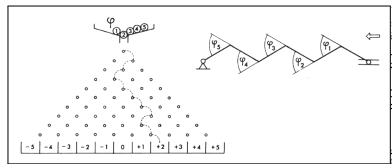
Figure 2: Evolution of a minimal drag body in a wind tunnel, starting from a randomly kinked plate.

¹I. Rechenberg: Cybernetic solution path of an experimental problem. *Royal Aircraft Establishment, Farnborough, Library Translation 1122*, 1965.

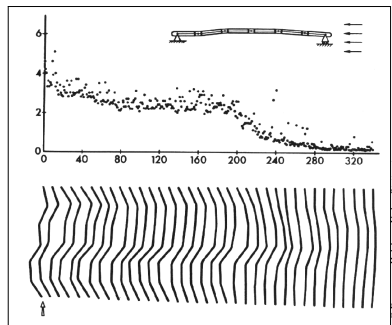
- six plates joint by five hinges
- hinges are adjustable, allowing for 51 angular positions
- size of search space: $51^5 = 345,025,251$
- (1 + 1)-ES used, mutation by Galton pin-board experiments

Galton's mutation scheme:

(reproduced from I. Rechenberg, 1973, p. 26, op. cit.)



Evolutionary dynamics:



Results (phenotypes):

(reproduced from I. Rechenberg, 1973, p. 28, op. cit.)

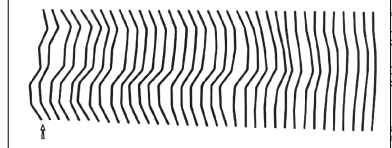
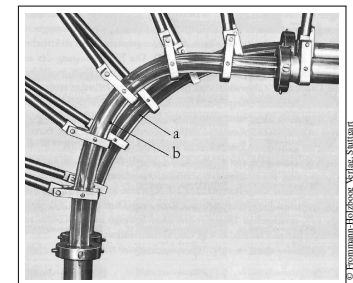
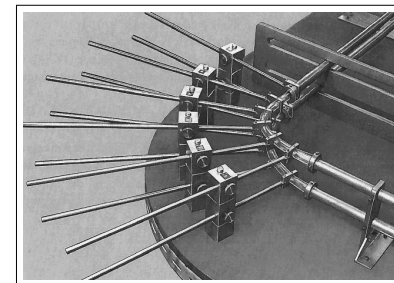


Figure 3: The *experimentum crucis* continued.

Evolution of a Pipe Bend

Find the shape of a 90 degree elbow such that flow resistance is minimal.



(reproduced from I. Rechenberg, 1973, p. 31, & 33, op. cit.)

Figure 4: Left: Experimental design using flexible plastic hoses. The lower hose is the (standard) reference design, the upper hose is for evolutionary optimization. Right: a – initial layout, b – final result obtained after about 300 generations.

☞ **industrial application:** optimized exhaust manifold

Evolution of a Two-Phase Nozzle by an Evolution Strategy²

Find the shape of a super-sonic two-phase nozzle that maximizes thrust.

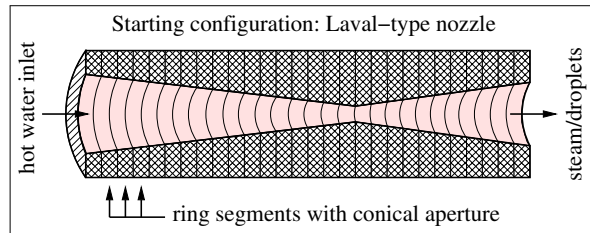


Figure 5: Hardware model of an *evolvable* nozzle.

- evolution with *variable* length genome
- segments (genes) can be duplicated and deleted
- diameters of aperture range from 6 to 36mm in 2mm steps, thickness of rings: 1cm

²J. Klockgether and H.-P. Schwefel: Two-phase nozzle and hollow core jet experiments. In D. G. Elliott, editor, *Proc. 11th Symp. Engineering Aspects of Magnetohydrodynamics*, pages 141–148, California Institute of Technology, Pasadena, CA, 24.–26. March, 1970.

- maximum length of nozzle 30cm
- more than $16^{30} \approx 10^{36}$ possible shapes

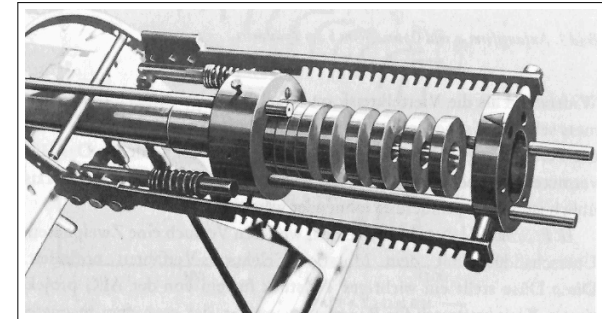
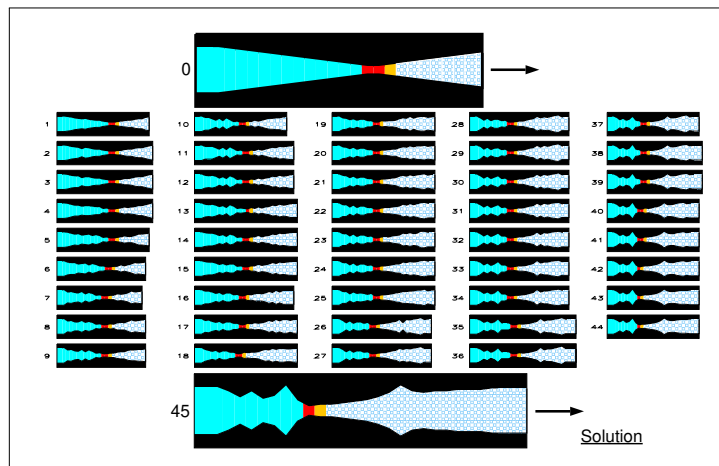


Figure 6: Experimental setup.³

- result after < 300 experiments in 14 days:
efficiency improvement from initially 55% to 80%

³Reproduced from I. Rechenberg [1], 1973, p. 34.

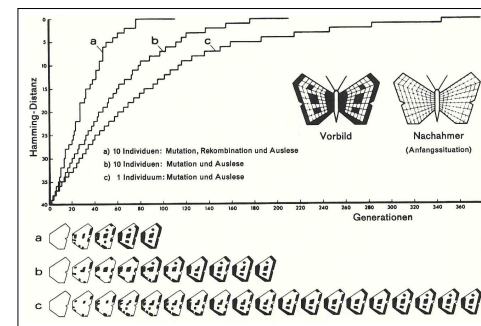


(adapted by M. Herdy from I. Rechenberg, 1973, p. 35, op. cit.)

Figure 7: Evolution and optimized shape of nozzle.

The Mimicry Experiment

- **Goal:** evolve a binary pattern (“Vorbild”) from scratch (“Nachahmer”)
- first example of applying the ES in binary $\mathbb{B}^N$ search spaces



(reproduced from I. Rechenberg, 1973, p. 85, op. cit.)

Figure 8: Performance comparison c: (1 + 1)-ES, b: (10 + 1)-ES, and a: (10/2 + 1)-ES.

- performance comparison of different ES including also $\mu/2$ -recombination
- probably *first* empirical evidence that recombination may have a performance advantage over simple mutative strategies on the OneMax problem
- Nowadays these ES operating in $\mathbb{B}^N$ are often called “EA” instead of “ES”, however, this is historically wrong!

A Short Recap of Evolution Strategies (ES)

Evolution Strategies⁴ are a class of Evolutionary Algorithms that use:

- ❶ mutation and recombination to generate λ offspring from μ parents
- ❷ perform truncation (aka breeding) selection denoted by (μ, λ) : only the μ best offspring individuals are selected as parents of the next generation, or $(\mu + \lambda)$: both the λ offspring and the μ parents are together object of selection to determine the parents of the next generation

Application domains (search spaces):

- discrete [4], combinatorial [5, 6], real-valued optimization, and mixtures [7, 8]
- **however, predominantly used in real-valued *unconstrained* optimization**
- popularized by the Covariance Matrix Adaptation (CMA) ES⁵

⁴See H.-G. Beyer [2]: [Scholarpedia: Evolution Strategies](#).

H.-G. Beyer and H.-P. Schwefel [3]: *Evolution Strategies: A Comprehensive Introduction*. Natural Computing 1(1):3–52, 2002.

⁵See N. Hansen, S.D. Müller, and P. Koumoutsakos [9]: *Reducing the Time Complexity of the Derandomized Evolution Strategy with Covariance Matrix Adaptation (CMA-ES)*.

Simple ES with Self-Adaptation and Recombination

$(\mu/\mu_I, \lambda)$ - σ SA-ES	line
Initialize $(\mathbf{y}_p, \sigma, \tau)$	1
Repeat	2
For $l := 1$ To λ	3
$\tilde{\sigma}_l := \sigma e^{\tau \mathcal{N}_l(0,1)}$	4
$\tilde{\mathbf{d}}_l := \mathcal{N}_l(\mathbf{0}, \mathbf{I})$	5
$\tilde{\mathbf{y}}_l := \mathbf{y}_p + \tilde{\sigma}_l \tilde{\mathbf{d}}_l$	6
$\tilde{f}_l := f(\tilde{\mathbf{y}}_l)$	7
$\tilde{\mathbf{a}}_l := (\tilde{f}_l, \tilde{\mathbf{y}}_l, \tilde{\sigma}_l)$	8
End	9
RankOffspringPopulation($\tilde{\mathbf{a}}_1, \dots, \tilde{\mathbf{a}}_\lambda$)	10
$\mathbf{y}_p := \langle \tilde{\mathbf{y}} \rangle$	11
$\sigma := \langle \tilde{\sigma} \rangle$	12
Until (Termination_Condition)	13
Return ($\mathbf{y}_p$)	14

- **Task:** optimize $f(\mathbf{y})$, where $\mathbf{y} \in \mathbb{R}^N$ (i.e., unconstrained)
- L3–9: produce λ offspring
- L4: mutate σ (mutation strength), $\tau = 1/\sqrt{2N}$
- L5: generate search direction
- L6: mutate parent by $\mathbf{w} = \sigma \tilde{\mathbf{d}}$
- L7: evaluate offspring
- L8: assemble offspring
- L11f: recombine the *best* μ offspring⁶ $\tilde{\mathbf{y}}$ and $\tilde{\sigma}$:

$$\langle \tilde{\mathbf{y}} \rangle := \frac{1}{\mu} \sum_{m=1}^{\mu} \tilde{\mathbf{y}}_{m;\lambda} \quad (1)$$

$$\langle \tilde{\sigma} \rangle := \frac{1}{\mu} \sum_{m=1}^{\mu} \tilde{\sigma}_{m;\lambda} \quad (2)$$

⁶“ $m; \lambda$ ” is the index of the m th best individual out of λ offspring (w.r.t. fitness).

$(\mu/\mu_I, \lambda)$ - σ SA-ES of Slide 14 Uses Isotropic Gaussian Mutations in $\mathbb{R}^N$

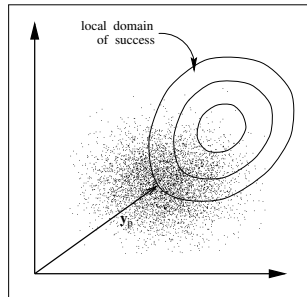


Figure 9: Isotropic Gaussian mutation samples in a 2-dimensional search space applied to a parental state $\mathbf{y}_p = \langle \tilde{\mathbf{y}} \rangle$ in Line 6, Slide 14.

- for “well shaped” local success domains, isotropic mutations $\mathbf{w} = \sigma \tilde{\mathbf{d}}$ suffice:
- $$\mathbf{w} \sim (\mathcal{N}(0, \sigma^2), \dots, \mathcal{N}(0, \sigma^2))^T = \sigma \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (3)$$
- probability density function:
- $$p(\mathbf{w}) = p(w_1, \dots, w_N) = \frac{1}{(\sqrt{2\pi}\sigma)^N} \exp\left(-\frac{w_1^2 + \dots + w_N^2}{2\sigma^2}\right) \quad (4)$$
- (hyper) surfaces of constant p are spherical shells
 - note, in high N -dimensional spaces the mutation vectors $\mathbf{w}$ are nearly located in the vicinity of a sphere of radius $\sigma\sqrt{N}$
 - mutation strength σ can be adapted by, e.g., $(\mu/\mu_I, \lambda)$ - σ SA-ES, Slide 14

Why Gaussian Mutations?

- ES design philosophy: you have to choose a mutation distribution, Fig. 3
- ☞ this is the main difference to Differential Evolution (DE) where the “distribution” is *implicitly* encoded in the parental population
- intuitively clear: smaller mutations should be more likely than larger
- ☞ uniform Monte-Carlo-Search does not make sense in high-dimensional search spaces and is excluded in unbounded search spaces such as the $\mathbb{R}^N$
- ☞ mutations have to sample the *relative* vicinity of the parents in order to learn the topology of the fitness landscape
- this implies that the mutation distribution is able to *adapt* to the fitness landscape
- assuming the existence of the *expected value* and the *variance* of the mutation distribution already fixes the distribution in the $\mathbb{R}^N$
- demanding maximum unbiasedness, i.e., maximum information content in the distribution the *maximum entropy principle* yields:
 - ☞ the Gaussian distribution for $\mathbb{R}^N$ search spaces and
 - ☞ the geometric distribution for $\mathbb{N}$ [10]

Why Intermediate Recombination? - The Genetic Repair

- recombination mixes the information of μ parents being the μ best offspring $\tilde{\mathbf{y}}_{m;\lambda}$ of the previous generation
- this is the main formal difference to mutation that acts on single parents
- intermediate μ/μ_I (multi-) recombination** takes the vector average of *all these* μ parent individuals yielding the parental centroid $\mathbf{y}_p$

$$\mathbf{y}_p := \langle \tilde{\mathbf{y}} \rangle_\mu, \quad \text{where} \quad \langle \tilde{\mathbf{y}} \rangle_\mu := \frac{1}{\mu} \sum_{m=1}^{\mu} \tilde{\mathbf{y}}_{m;\lambda} \quad (5)$$

Why can this be beneficial?

- two hypotheses:
 - Building Block Hypothesis (BBH, GOLDBERG) [11]:**
in discrete recombination “good genetic material” flows from the different parents into the offspring
 - Genetic Repair (GR) Principle (BEYER) [12]:**
recombination extracts *similarities* from the parents and suppresses harmful parts of the parents
- both hypotheses can be correct, the benefits of recombination depend strongly on the fitness function to be optimized

On the Benefits of Recombination - The GR-Effect

- consider the $(\mu/\mu_I, \lambda)$ -ES
- consider evolution of the parent centroid $\mathbf{y}_p$ in Line 11, Slide 14, and its fitness $f(\mathbf{y}_p)$
- using Eq. (5) one sees

$$\mathbf{y}_p^{(g+1)} := \langle \tilde{\mathbf{y}}^{(g+1)} \rangle_\mu \quad (6)$$

- offspring $\tilde{\mathbf{y}}_l$ are obtained from the parental centroid $\mathbf{y}_p^{(g)}$ by applying mutations $\mathbf{w}_l$

$$\tilde{\mathbf{y}}_l^{(g+1)} := \mathbf{y}_p^{(g)} + \mathbf{w}_l \quad (7)$$

- inserting (7) in (6) one gets

$$\mathbf{y}_p^{(g+1)} = \langle \mathbf{y}_p^{(g)} + \mathbf{w} \rangle_\mu = \mathbf{y}_p^{(g)} + \langle \mathbf{w} \rangle_\mu \quad (8)$$

- now, take a geometrical perspective . . .

A Geometrical Explanation of the Genetic Repair (GR) Effect

consider parental centroids of two consecutive generations of a $(\mu/\mu_I, \lambda)$ -ES in search space:

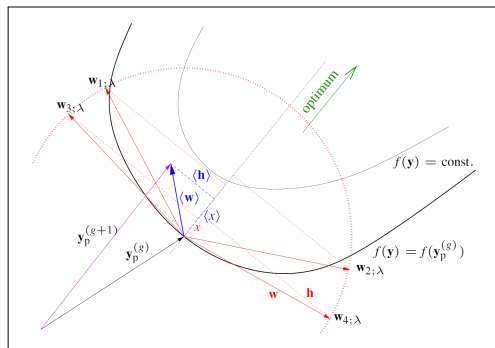


Figure 10: The Genetic Repair effect visualized in a $(4/4_I, 10)$ -ES.

- fitness of parental centroid at (g) defines local success region
- mutation $\mathbf{w}$ can be split in a component x in optimum direction and a perpendicular part $\mathbf{h}$

$$\mathbf{w} = x\mathbf{e}_{\text{opt}} + \mathbf{h} \quad (9)$$

- selected mutations have tendency into (local) optimum direction
- $\mathbf{h}$ is almost selectively neutral and uncorrelated
- averaging (center of mass) results in reduced $\langle \mathbf{h} \rangle$

- $\mathbf{h}$ of mutation $\mathbf{w}$ is harmful part (it drives away the mutant from the optimum direction)
- ➔ **recombination** reduces, i.e., it “*repairs*” the harmful influence on the recombinant:

The Genetic Repair Principle (GR)

INTERMEDIATE RECOMBINATION = STATISTICAL ERROR CORRECTION = “GENETIC REPAIR”

Equivalent Interpretations:

- recombination *correlates* the *selected* parents
- ➔ thus, it *extracts similarities* or *commonalities* from the parents
- note:** this is *in opposition to building block hypothesis*
- explanations hold also for dominant (discrete) recombination; however, GR is statistically performed and mutations are transformed into “genetic variation” building a *species-like* parental distribution crowded around a “wild type” (i.e., average individual)

Is There Genetic Repair in Binary Search Spaces?

... at least similarity extraction:

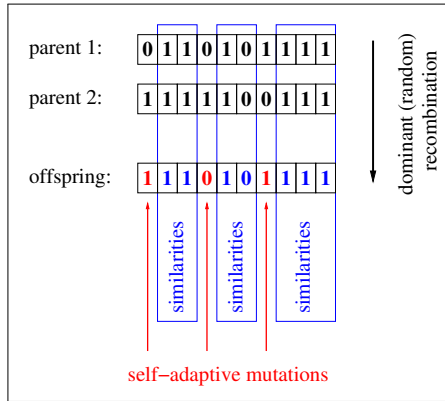


Figure 11: On the interpretation of dominant recombination in binary $\mathbb{B}^N$ search spaces.

- parents share similarities since they are selected w.r.t. their fitness
- these similarities are conserved by the coordinate-wise random transfer of dominant recombination
- due to the random sampling, those parental components that do not agree are randomly set
- this may be interpreted as a self-adaptive mutation

An Alternative in $\mathbb{R}^N$ and $\mathbb{Z}^N$: Mutate Large, But Inherit Small?

- evolutionary progress is always the result of two antagonists:

$$\frac{\text{EVOLUTIONARY PROGRESS}}{\text{PROGRESS}} = \frac{\text{PROGRESS GAIN}}{\text{PROGRESS GAIN}} - \frac{\text{PROGRESS LOSS}}{\text{PROGRESS LOSS}}$$

- intermediate recombination in Fig. 10 reduces the loss part **h**
- thus, the length of the mutations can be increased resulting in an increased gain part **x**
- as proposed by OSTERMEIER⁷ and analyzed in [14, 15], a similar effect can be obtained by *rescaled mutations* leading to “mutate large, inherit small”

$$\mathbf{y}_p^{(g+1)} := \mathbf{y}_p^{(g)} + \frac{1}{\kappa} \langle \mathbf{w} \rangle, \quad \kappa \geq 1. \quad (10)$$

- CAUWET & TEYTAUD [16] have shown that this principle can deliver the maximum theoretically possible performance in *noisy optimization*

⁷According to Rechenberg [13, p. 195].

Design Principles - ES Philosophy

- 1 use *natural problem representation* (if possible), avoid genotype-phenotype mappings
- 2 ES is based on the variation-selection-principle
- 3 variation should produce more-or-less *unbiased* individuals (undirected in the search space)
- 4 selection provides the direction of the evolution based on the objectives to be optimized
- 5 selection is by truncation (breeder selection), either as (μ, λ) or $(\mu + \lambda)$
- 6 as for $(\mu/\mu_t, \lambda)$ -ES the truncation ratio should be in the intervall $\mu/\lambda = 1/4 \dots 1/2$
- 7 the selection process can even be “subjective”
- 8 in combinatorial and binary search spaces use plus-selection
- 9 in real-valued search spaces with noisy (i.e., stochastic) or time-dependent objective functions, comma-selection is recommended

- 10 **variation** is by:
 - 1 mutation, i.e., changing a single individual
 - 2 recombination, i.e., combining two or more individuals
- 11 **variation operators** should be problem-specifically designed depending on the search spaces (real-valued, binary, combinatorial, etc.)
- 12 **mutation operators** should
 - 1 be consistent, i.e. do not produce infeasible individuals
 - 2 be unbiased, if possible (maximum entropy design principle)
 - 3 be scaleable, i.e. should allow for a strategy parameter that controls the mutation strength, if possible
 - 4 allow for correlated mutations
- 13 **mutation strength or probability** (if it can be defined) should be controlled by:
 - 1 1/5-rule, or more generally 1/s-rule, in conjunction with plus-selection
 - 2 self-adaptation
 - 3 advanced control paradigms such as cumulative step size adaptation (CSA) or Meta-ES

14 recombination operators should

- 1 be consistent, i.e., do not produce infeasible individuals
- 2 perform genetic repair, i.e., extracting or emphasizing similarities of the individuals
- 3 respect the maximum entropy principle
- 4 be done in terms of multi-recombination, such as in $(\mu/\mu_I, \lambda)$ -ES

15 population sizing may depend on the dimensionality N of the search space, here are some rules of thumb for $\mathbb{R}^N$ search spaces:

- 1 for simple optimization problems or expensive objective functions (long running time to evaluate an individual) use $\lambda = 4 + \lfloor 3 \ln(N) \rfloor$ or $\lambda = \lceil e\sqrt[3]{N} \rceil$
- 2 finding global optima on highly multimodal objective functions may require restarts and/or an increased population size that can scale proportional to $\sqrt{N}$, $\sqrt{N} \ln(N)$, N , or even higher (see GECCO'23 paper [17]).
- 3 getting arbitrarily close to the optimizer in the case of objective functions disturbed by strong noise may even require an exponentially increasing population size.⁸

⁸See M. Hellwig and H.-G. Beyer [18]: *Evolution under Strong Noise: A Self-Adaptive Evolution Strategy Can Reach the Lower Performance Bound - the pcMSA-ES*. Parallel Problem Solving from Nature, PPSN XIV, pp. 26–36, Springer, 2016, see also [19, 20].

Evolutionary Gradient Search and “Natural ES”

What are Gradient (Based) Strategies?

If one wants to minimize a function $f(\mathbf{y})$, $\mathbf{y} \in \mathbb{R}^N$

Iterative scheme:

$$\mathbf{y}^{(g+1)} = \mathbf{y}^{(g)} - \eta^{(g)} \nabla f(\mathbf{y}^{(g)}) \quad (11)$$

or more general

$$\mathbf{y}^{(g+1)} = \mathbf{y}^{(g)} - \eta^{(g)} \mathbf{C}^{(g)} \nabla f(\mathbf{y}^{(g)}) \quad (12)$$

as long as $\mathbf{C}^{(g)} \in \mathbb{R}^{N \times N}$ is *positive definite*

SALIMANS ET AL. [21] used normally distributed mutations $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ and *called* this

$$\mathbf{y}^{(g+1)} = \mathbf{y}^{(g)} - \alpha \sum_{i=1}^{\lambda} f(\mathbf{y}^{(g)} + \mathbf{w}_i) \mathbf{w}_i \quad (13)$$

update scheme “Evolution Strategy” (with *wrong* reference to RECHENBERG)

What is the meaning of: $\sum_{i=1}^{\lambda} f(\mathbf{y} + \mathbf{w}_i) \mathbf{w}_i$?

Since in high-dimensional spaces the (expected) length of $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ is

$$\mathbb{E}[\|\mathbf{w}_i\|] \simeq \sigma \sqrt{N} \quad (14)$$

this is a Monte Carlo estimator of a *surface integral* in $\mathbb{R}^N$

$$\sum_{i=1}^{\lambda} f(\mathbf{y} + \mathbf{w}_i) \mathbf{w}_i \simeq \oint_{\mathbf{x} \in \partial V} f(\mathbf{y} + \mathbf{x}) d\mathbf{A}(\mathbf{x}) \quad (15)$$

Applying Gauss’ Theorem: $\oint_{\partial V} f(\mathbf{x}) d\mathbf{A} = \iiint_V \nabla f dV$ and

divide by the volume V of the ball and taking the limit $V \rightarrow 0$, i.e., $\sigma \rightarrow 0$

$$\lim_{V \rightarrow 0} \frac{1}{V} \sum_{i=1}^{\lambda} f(\mathbf{y} + \mathbf{w}_i) \mathbf{w}_i \simeq \lim_{V \rightarrow 0} \frac{1}{V} \oint_{\partial V} f(\mathbf{y} + \mathbf{x}) d\mathbf{A}(\mathbf{x}) = \lim_{V \rightarrow 0} \frac{1}{V} \iiint_V \nabla f dV = \nabla f \quad (16)$$

one recovers the *coordinate-free* definition of the gradient!

☞ SALIMANS ET AL. “Evolution Strategy” is a vanilla gradient strategy!

☞ ... and this is not SALIMANS’ ET AL. invention, but was already proposed by R. SALOMON in the late 1990s as “Evolutionary Gradient Search” [22]

Natural Evolution Strategies (NES)

- approach developed by the SCHMIDHUBER group in the late 2000s in terms of so-called “Natural Evolution Strategies” [23]
- also used in terms of the Information Geometric Optimization (IGO) “philosophy” by OLLIVIER ET AL. [24] to propose a principled foundation of CMA-ES, see also AKIMOTO ET AL. [25]

Philosophy of the NES approach

- consider ES as probabilistic optimization procedures that evolve sampling distribution(s) with pdf $p(\mathbf{y}|\boldsymbol{\theta})$ towards the optimizer $\hat{\mathbf{y}}$ of $f(\mathbf{y})$
- evolution is controlled through the $\boldsymbol{\theta}$ parameter set of the probability distribution family, e.g. Gaussian: $\mathbf{y} \sim \mathcal{N}(\bar{\mathbf{y}}, \boldsymbol{\Sigma}) \Rightarrow \boldsymbol{\theta} = (\bar{\mathbf{y}}, \boldsymbol{\Sigma})$, cf. (43)
- it is the aim to maximize (minimize) the expected value of $f(\mathbf{y})$, i.e.

$$E_f(\boldsymbol{\theta}) := \mathbb{E}[f|\boldsymbol{\theta}] = \int f(\mathbf{y}) p(\mathbf{y}|\boldsymbol{\theta}) d^N \mathbf{y} \quad (17)$$

- Eq. (17) defines a *potential* and the gradient $\nabla_{\boldsymbol{\theta}} E$ determines the $\boldsymbol{\theta}$ flow
- NB, $p(\mathbf{y}|\boldsymbol{\theta})$ must be controllable via $\boldsymbol{\theta}$ to represent $\hat{\mathbf{y}}$ by concentrating the probability mass at $\hat{\mathbf{y}}$

- gradient descent (ascent) gives rise to an ordinary differential equation (ODE) system

$$\frac{d\theta}{dt} = -\nabla_{\theta} E_f(\theta)|_{\theta=\theta(t)} \quad (18)$$

- the idea is to numerically *approximate* the θ flow by
 1. discretizing the time (thus, getting an iterative scheme, e.g. Euler method)

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} E_f(\theta)|_{\theta=\theta^{(t)}} \quad (19)$$

2. approximating the gradient $\nabla_{\theta} E_f$ by Monte-Carlo sampling (yielding an evolutionary algorithm)

$$\nabla_{\theta} E_f(\theta) \approx \frac{1}{\lambda} \sum_{i=1}^{\lambda} f(y_i) \nabla_{\theta} \ln p(y_i|\theta), \quad \text{where } y_i \sim \mathcal{N}(\bar{y}, \Sigma) \quad (20)$$

- unfortunately, the resulting algorithms may be instable w.r.t. Σ matrix evolution resulting in Σ changes too fast, GLASMACHERS [26], BEYER [27]
- i.e., the apparent information gain of Σ is unjustifiable
- ➔ SCHMIDHUBER's group [23] proposed the use of “*natural gradients*”

What is the “*natural gradient*”?

- introduced by AMARI [28] (1989) in context of *information geometry*
- Task: find the direction in θ -space that maximally improves E_f while keeping the *information gain* I at a small (fixed) ϵ level
- information gain is measured by *relative entropy* RENYI [29] (1961), aka KL (KULLBACK-LEIBLER) divergence

$$I(P(\theta + \delta\theta)|P(\theta)) = \int \ln \left(\frac{p(y|\theta + \delta\theta)}{p(y|\theta)} \right) p(y|\theta + \delta\theta) d^N y \quad (21)$$

- this leads to a constrained optimization problem

$$E_f(\theta + \delta\theta) - E_f(\theta) = \sum_i \frac{\partial E_f(\theta)}{\partial \theta_i} \delta\theta_i + \dots \rightarrow \text{Max!} \quad (22a)$$

$$\text{s.t. } I(P(\theta + \delta\theta)|P(\theta)) = \epsilon \quad (22b)$$

for determining the direction $\delta\theta$

- the solution of which is given by

$$\delta\hat{\theta} = \sqrt{\frac{2\epsilon}{\nabla_{\theta}^T E_f(\theta) \mathcal{I}^{-1}(\theta) \nabla_{\theta} E_f(\theta)}} \mathcal{I}^{-1}(\theta) \nabla_{\theta} E_f(\theta) \quad (23)$$

- where $\mathcal{I}^{-1}(\theta)$ is the inverse of FISHER's information matrix [30]

$$(\mathcal{I}(\theta))_{ij} := \int \frac{\partial p(y|\theta)}{\partial \theta_i} \frac{\partial p(y|\theta)}{\partial \theta_j} p(y|\theta) d^N y \quad (24)$$

- AMARI defines the “*natural gradient*” as

$$\tilde{\nabla}_{\theta} E_f(\theta) := \mathcal{I}^{-1}(\theta) \nabla_{\theta} E_f(\theta) \quad (25)$$

- replacing ∇_{θ} by $\tilde{\nabla}_{\theta}$ in (18), one gets the ODE

$$\frac{d\theta}{dt} = -\mathcal{I}^{-1}(\theta) \nabla_{\theta} E_f(\theta)|_{\theta=\theta(t)} \quad (26)$$

being the starting point of NES

Natural Evolution Strategies (NES)

- using (26), one gets instead of (19)

$$\theta^{(t+1)} = \theta^{(t)} - \eta \mathcal{I}^{-1}(\theta) \nabla_{\theta} E_f(\theta)|_{\theta=\theta(t)} \quad (27)$$

- calculating $\mathcal{I}^{-1}(\theta)$ explicitly is almost always excluded due to the dimensionality of $\mathcal{I}$ ⁹
- however, if Gaussian mutation distribution is suitably parameterized, it is possible to bypass $\mathcal{I}^{-1}(\theta)$ and to get $\mathcal{I}^{-1}(\theta) \nabla_{\theta} E_f(\theta)$ directly, leading to the *exponential* NES (xNES) [26]

⁹E.g., if θ comprises $\mathcal{O}(N^2)$ free parameters, $\mathcal{I}$ is an $\mathcal{O}(N^2 \times N^2)$ matrix and calculating $\mathcal{I}^{-1}$ is of $\mathcal{O}(N^6)$.

How do NES Relate to Classical ES?

- a straight implementation of (27) or (26) does not yield competitive NES algorithms:
- as has been shown by BEYER [27], (26) applied to a general quadratic model $f(\mathbf{y}) = \mathbf{a}^T \mathbf{y} - \mathbf{y}^T \mathbf{Q} \mathbf{y}$ yields *provably* a disappointing *sublinear convergence order*
- ☞ the natural gradient descent (ascent) does not automatically yield competitive algorithms
- there are two ingredients (actually implemented in xNES and other NES), which cannot be derived from (27), that are responsible for a linear convergence order:
 - ☞ f must be replaced by a time (iteration) dependent utility function [27], this is a *conditio sine qua non*
 - ▶ additionally one finds component-group wise learning rates in NES instead of a single η
 - ☞ this means that competitive NES *do not follow* the natural gradient flow dictated by (26)

On the Search Behavior of ES in $\mathbb{R}^N$

How does the ES explore the search space?

- often used picture: Population traces the gradient path
- this is based on the following observations
 - 1 ES exhibits linear convergence order just like classical gradient strategies
 - 2 Claims in publications:
 - ★ “Evolution strategies (ES) can be best described as a gradient descent method which uses gradients estimated from stochastic perturbations around the current parameter value.”¹⁰
 - ★ “. . . instead of computing the exact gradient, ES computes an approximation from all the sample points (called pseudo-offspring) generated from parent”¹¹
- NB: This is due to a misleading statement in a paper by Salimans et al. [21]: Evolution Strategies as a Scalable Alternative to Reinforcement Learning.¹²

¹⁰ <https://www.inference.vc/evolutionary-strategies-embarrassingly-parallelizable-optimization/>

¹¹ X. Zhang, J. Clune, and K.O. Stanley: On the Relationship Between the OpenAI Evolution Strategy and Stochastic Gradient Descent. ArXiv e-prints, abs/1712.06564

¹² T. Salimans, J. Ho, X. Chen, S. Sidor, and I. Sutskever. ArXiv e-prints, abs/1703.03864

Experimental “Evidence” of Gradient Descent Picture

- How to visualize the search behavior in $\mathbb{R}^N$?
- if one projects N -dimensional individual $\mathbf{y} := (y_1, \dots, y_N)^T$ into (x_1, x_2) -plane using (RECHENBERG, [13, p. 75])

$$x_1 := \sqrt{y_1^2 + \dots + y_{(N/2)}^2}, \quad x_2 := \sqrt{y_{(N/2)+1}^2 + \dots + y_N^2}, \quad (28)$$

one observes indeed some kind of “gradient diffusion”

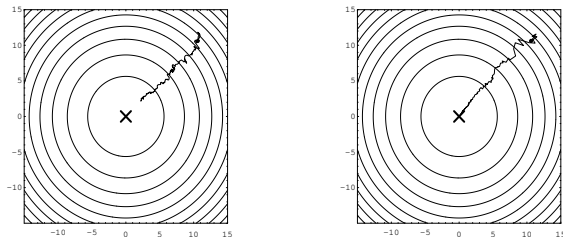


Figure 12: Path of the best individual in a $(4, 20)$ -ES (left) and a $(4/4, 20)$ -ES (right) on the $N = 100$ -dimensional sphere model after Projection (28) into 2D over 200 generations. “•”: start, “x”: optimizer.

- Fig. 12 presents a strong support for the gradient diffusion picture, however
- ⇒ What would be the use of ES at all?
- ⇒ probability of leaving local attractors would be very small
- ⇒ one should better use multi-start gradient strategies

Is this the real picture of the search behavior of ES?

- No, Projection (28) is misleading:
- lumping together $N/2$ components $\Rightarrow$ central limit theorem of statistics dampens the variance of the random components by a factor of $2/N$
- behavior of single components of the $\mathbf{y}$ vector is not correctly reflected
- ☞ single components of $\mathbf{y}$ must be considered

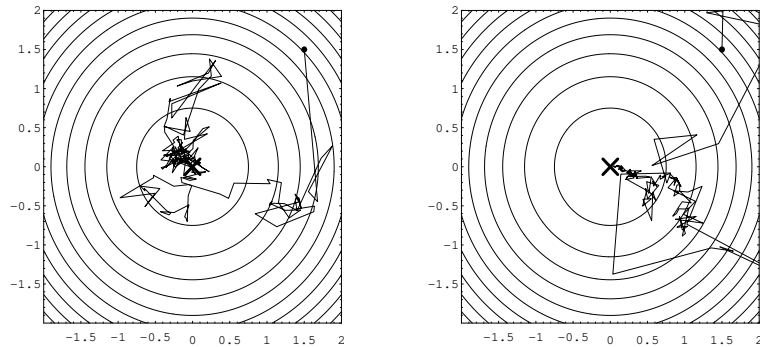


Figure 13: The $x_1 := y_1$ and $x_2 := y_2$ components (x_1 horizontal axis, x_2 vertical axis) of the evolution path of the best individual of the ES runs of Fig. 12, Slide 35 are displayed. Left: (4, 20)-ES, right: (4/4, 20)-ES.

- actually realized evolution path is much more random as can be seen on Slide 37
- however, this random walk is restricted by selection
- approach to the optimizer $\Leftrightarrow$ EXPLOITATIVE POWER of the ES
- can be described by **Evolutionary Progress Principle (EPP)**, cf. Slide 22
- note, concrete form of EPP depends on the definition of “progress”
- however, it is always related to a decomposition of the mutation vector $\mathbf{w}$ or the vector describing the change of the parental centroid from $\mathbf{g}$ to $\mathbf{g} + 1$, cf. Fig. 10, Slide 19
- **general observation:**

$$\begin{cases} \text{gain part} & \Leftrightarrow x\text{-component} & \Leftrightarrow \text{EXPLOITATION} \\ \text{loss part} & \Leftrightarrow \mathbf{h}\text{-vector} & \Leftrightarrow \text{EXPLORATION} \end{cases} \quad (29)$$

Q: How to quantify Exploitation/Exploration? ¹³

¹³ See H.-G. Beyer [31]: On the “Explorative Power” of ES/EP-like Algorithms.

DOI: 10.1007/BFB0040785

Different options to define the exploitation/exploration ratio

- 1 decomposition of the expected value of the parental centroid change $\langle \mathbf{y} \rangle^{(g)} - \langle \mathbf{y} \rangle^{(g+1)}$ according to

$$\frac{\text{Exploitation}}{\text{Exploration}} := \frac{\mathbb{E}[R - \tilde{R}]}{\mathbb{E}[\|\mathbf{h}\|]} = \frac{\varphi}{\mathbb{E}[\|\mathbf{h}\|]} \quad (30)$$

- 2 relating the fictive length of the expected change in local gradient direction to the perpendicular part (perpendicular w.r.t. the local gradient) of the parental centroid change

► fictive length is also referred to as *normal progress* φ_R

$$\varphi_R = \frac{\bar{Q}}{\|\nabla f(\mathbf{y}_p)\|}, \quad \bar{Q} - \text{QUALITY GAIN}, \quad \mathbf{y}_p = \langle \mathbf{y} \rangle^{(g)} \quad (31)$$

► where quality gain is defined by

$$\bar{Q} = \mathbb{E} \left[f(\langle \mathbf{y} \rangle^{(g+1)}) - f(\mathbf{y}_p) \right] \quad (32)$$

► and the exploitation/exploration ratio reads

$$\frac{\text{Exploitation}}{\text{Exploration}} := \frac{\varphi_R}{\mathbb{E}[\|\mathbf{h}\|]} \quad (33)$$

$$\begin{cases} \text{progress in local gradient direction} & \Leftrightarrow \text{EXPLOITATION} \\ \text{perpendicular part} & \Leftrightarrow \text{EXPLORATION} \end{cases} \quad (34)$$

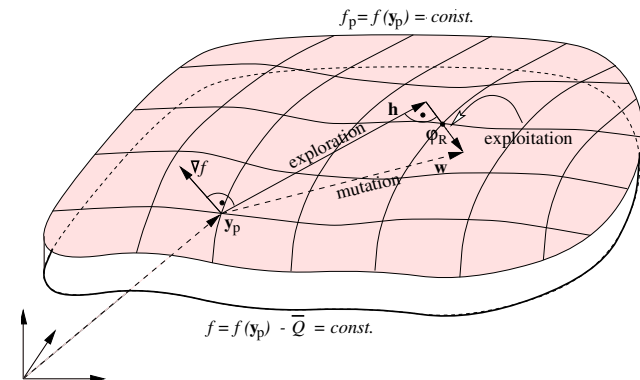


Figure 14: Visualization of exploration vs. exploitation based on normal progress. The surface displayed represents equal function values (i.e., $\mathbf{y} \in \mathbb{R}^3$).

Asymptotic $N \rightarrow \infty$ exploration-exploitation behavior (sphere model)

- isotropic Gaussian mutations: $E[||\mathbf{h}||] \simeq \sigma\sqrt{N}$
- as for $(\mu/\mu_I, \lambda)$ -ES on sphere model, Definition (29) yields

$$\max[\varphi_{\mu/\mu, \lambda}] \simeq \frac{R}{N} \mu \frac{c_{\mu/\mu, \lambda}^2}{2} \Leftrightarrow \sigma = \mu c_{\mu/\mu, \lambda} \frac{R}{N}$$

and

$$E[||\mathbf{h}_{\mu/\mu, \lambda}||] \simeq \frac{R}{N} \mu c_{\mu/\mu, \lambda} \sqrt{N}$$

- thus

$$\boxed{\frac{\text{Exploitation}}{\text{Exploration}} \simeq \left(\frac{1}{\sqrt{N}} \right)} \quad (35)$$

- this also holds for each single mutation

Lessons learned

- 1 **Exploitation:** ability of an ES to evolve into a desirable progress direction
 ↳ acts locally in one dimension
- 2 **Exploration:** process that drives the offspring away from the local progress direction
 ↳ *random walk* on an $(N - 1)$ -dimensional manifold, locally perpendicular to local progress direction
- 3 actual “path” of the population in search space does *not* follow the local gradient
- 4 **Are ESs path-oriented search methods?**
 ↳ Yes, Brownian random path
- 5 actual “path” of population in search space is reminiscent of *serpentes* in mountainous regions

While a single ES run performs a restricted random walk and not a gradient descent/ascent the question still remains:

What is the Average Search Behavior?¹⁴

- this is the question w.r.t. the mean value dynamics describing the approach to the optimizer
- this is closely related to theoretical and empirical performance analyses:
 - ▶ given an objective function model $f(\mathbf{y})$ to be optimized, how fast does the ES approach the optimizer?
 - ▶ how is the influence of the model parameters (e.g., condition number) on the ES performance?
 - ▶ convergence order and computational resources needed
 - ▶ ideally, we want to calculate the dynamics describing the approach towards the optimizer
 - ▶ getting information how strategy specific parameters (e.g. population size, truncation ratio) influence the performance

Ellipsoid Model Used: $f(\mathbf{y}) = \sum_{i=1}^N a_i y_i^2, \quad a_i > 0 \quad (36)$

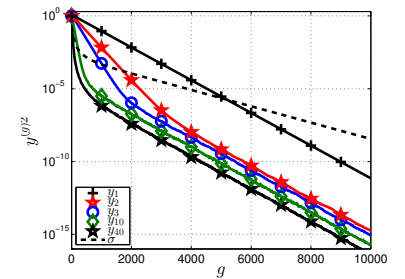
¹⁴H.-G. Beyer and A. Melkozerov [32]: The Dynamics of Self-Adaptive Multi-Recombinant Evolution Strategies on the General Ellipsoid Model. *IEEE Transactions on Evolutionary Computation*, 2014. DOI: 10.1109/TEVC.2013.2283968.

Mean Value Dynamics of the $(\mu/\mu_I, \lambda)$ - σ SA-ES

```

1  $\sigma^{(0)} := \sigma_{init}; \mathbf{y}^{(0)} := \mathbf{y}_{init}$ 
2  $g := 0$ 
3 Repeat
4   For  $l = 1, \dots, \lambda$ 
5      $\tilde{\sigma}_l := \sigma^{(g)} e^{\tau \mathcal{N}(0,1)}$ 
6      $\mathbf{d}_l := \mathcal{N}_I(\mathbf{0}, \mathbf{I})$ 
7      $\tilde{\mathbf{y}}_l := \mathbf{y}^{(g)} + \tilde{\sigma}_l \mathbf{d}_l$ 
8      $\tilde{f}_l := f(\tilde{\mathbf{y}}_l)$ 
9   End
10   $\tilde{\mathbf{f}}_{sort} := \text{sort}(\tilde{f}_{1 \dots \lambda})$ 
11   $\sigma^{(g+1)} := \frac{1}{\mu} \sum_{m=1}^{\mu} \tilde{\sigma}_{m;\lambda}$ 
12   $\mathbf{y}^{(g+1)} := \frac{1}{\mu} \sum_{m=1}^{\mu} \tilde{\mathbf{y}}_{m;\lambda}$ 
13   $g := g + 1$ 
14 Until termination

```

Figure 15: $(\mu/\mu_I, \lambda)$ - σ SA-ES**Figure 16:** Dynamics of the $(3/3_I, 10)$ -ES on a fitness function (36) with $a_i = i$ and $N = 40$. The quadratic deviation of y_i from the optimizer is displayed for the components $i = 1, 2, 3, 10, 40$. Additionally, the mutation strength σ has been plotted. ES learning parameter: $\tau = 1/\sqrt{N}$. The graphs are averages over 1000 independent runs.

- Fig. 16 shows that the squared y_i -components approach exponentially fast the optimizer

remarkable: after a transient phase, the slope of the y_i^2 are the same

- the steady state solution reads:

$$(y_i^{(g)})^2 = b_i e^{-\nu g}, \quad b_i > 0, \nu > 0 \quad (37)$$

$$\sigma^{(g)} = \sigma_0 e^{-\frac{\nu}{2} g}, \quad \sigma_0 > 0 \quad (38)$$

note, this already implies *linear convergence order*

- $\nu > 0$ is the smallest eigenvalue of an eigenvalue problem the asymptotic solution of which is [32]

$$\nu \simeq 2\sigma_{ss}^* c_{\mu/\mu, \lambda} \frac{\min(a_i)}{\sum_{k=1}^N a_k} \quad (39)$$

- Eq. (37) yields the important message that ...

Non-Isotropic Mutations in $\mathbb{R}^N$

1. Non-correlated Independently Distributed Gaussian Mutations

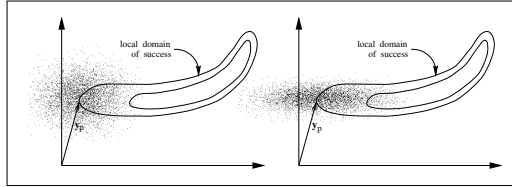


Figure 18: Success domains with preference directions parallel to certain coordinate directions are better treated by Gaussian mutation vectors the components of which have different mutations strengths (lhs: isotropic, rhs: ellipsoidal mutations).

$$\mathbf{w} \sim (\mathcal{N}(0, \sigma_1^2), \dots, \mathcal{N}(0, \sigma_N^2))^T, \quad p(\mathbf{w}) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{w_i^2}{2\sigma_i^2}\right) \quad (40)$$

- there is a set of N strategy parameters σ_i to be learned

$$(\sigma_1, \dots, \sigma_i, \dots, \sigma_N)^T \quad (41)$$

- (hyper) surfaces of constant p are axes-parallel ellipsoids

ES mean value dynamics *does not* follow the gradient of $f(\mathbf{y})$

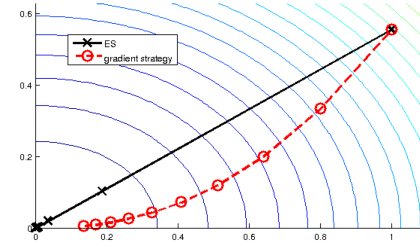


Figure 17: In the steady state, the ES follows in expectation a straight line towards the optimizer when applied to quadratic objective functions.

In the steady state:

- the ES does not follow the gradient path *even not on average* (exception: sphere, due to symmetry)
- instead the ES approximates on average the Newton-direction even though *only isotropic mutations* are used
- this also holds for cumulative step-size adaptation (CSA) [33]

2. Correlated Gaussian Mutations

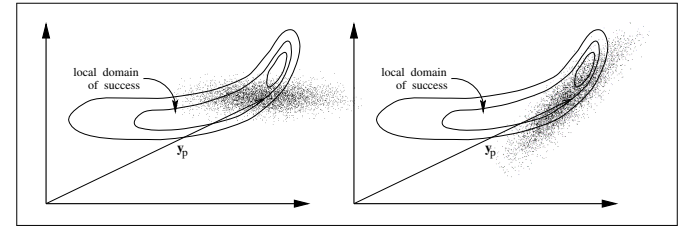


Figure 19: *Rotated* mutation ellipsoids (rhs) are better suited for the parent y_p .

- correlated* $\mathbf{w}$ mutations are to be used to obtain mutation ellipsoids arbitrarily oriented in search space

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \Sigma) \quad (42)$$

$$p(\mathbf{w}) = \frac{1}{(\sqrt{2\pi})^N} \frac{1}{\sqrt{\det[\Sigma]}} \exp\left(-\frac{1}{2} \mathbf{w}^T \Sigma^{-1} \mathbf{w}\right) \quad (43)$$

- Σ is the *covariance matrix* and Σ^{-1} its inverse

Remarks

- covariance matrix Σ contains $N(N+1)/2$ independent parameters to be learned (because Σ is symmetric)
- ➔ there are N object parameters to be evolved in order to optimize $f(\mathbf{y})$, *but*, there are $N(N+1)/2$ Σ -matrix components to be learned, too!
- ➔ using an ES that *learns* correlated mutations via covariance matrix Σ makes sense only when one has a budget of function evaluations that is greater than kN^2

How to generate correlated mutations?

- correlated mutations $\mathbf{w}$ can be produced by linear transformation of iid standard normally distributed vectors $\mathbf{z} = (\mathcal{N}_1(0, 1), \dots, \mathcal{N}_N(0, 1))^T$ by a two-step process
 - 1 calculating the direction: $\mathbf{d} := \mathbf{M}\mathbf{z}$
 - 2 scaling the length: $\mathbf{w} := \sigma\mathbf{d}$
- since $E[\mathbf{w}] = E[\sigma\mathbf{M}\mathbf{z}] = \sigma\mathbf{M}E[\mathbf{z}] = \mathbf{0}$, one finds using the definition of Σ

$$\Sigma = E[\mathbf{w}\mathbf{w}^T] = \sigma^2 E[\mathbf{M}\mathbf{z}\mathbf{z}^T\mathbf{M}^T] = \sigma^2 \mathbf{M}E[\mathbf{z}\mathbf{z}^T]\mathbf{M}^T = \sigma^2 \mathbf{M}\mathbf{M}^T \quad (44)$$

The Matrix Adaptation Idea

How can one get $\mathbf{M}$ if Σ were known?

- ☞ take the “square root” of Σ in (44), i.e., $\mathbf{M} = \frac{1}{\sigma}\sqrt{\Sigma}$; this can be done by:
 - ▶ CHOLESKY-decomposition
 - ▶ matrix square root via eigenvalue decomposition
- ☞ these operations are computationally demanding for large N

However, how can one get the covariance matrix Σ ?

- it must be derived from the evolutionary dynamics of the real ES run
- this is what the Covariance Matrix Adaptation (CMA) ES [34] does

But, why not deriving the $\mathbf{M}$ matrix from the observed ES dynamics directly?

- analysis of the CMA-ES revealed that one can (approximately) rewrite the CMA-ES algorithm and remove its “C”
- as a result one obtains very simple Matrix Adaptation (MA) ES that perform equally well as the CMA-ES¹⁵

☞ **C-related numerical operations are no longer needed!**

¹⁵See H.-G. Beyer and B. Sendhoff [35]: Simplify Your Covariance Matrix Adaptation Evolution Strategy. *IEEE Transactions on Evolutionary Computation* 21(5):746–759, 2017. DOI: 10.1109/TEVC.2017.2680320

A Simple Recombinative MA-ES with Self-Adaptation (SA)

$(\mu/\mu_l, \lambda)$ - σ SA-MA-ES	line	
Initialize $(\mathbf{y}, \sigma, \tau, \tau_M, \mathbf{M} := \mathbf{I})$	1	• L3–10: produce λ offspring
Repeat	2	• L4: mutate σ (mutation strength), $\tau := 1/\sqrt{2N}$
For $l := 1$ To λ	3	• L5f: generate search direction
$\tilde{\sigma}_l := \sigma e^{\tau \mathcal{N}_l(0, 1)}$	4	• L7: mutate parent by $\mathbf{w} = \tilde{\sigma} \tilde{\mathbf{d}}$
$\tilde{\mathbf{z}}_l := \mathcal{N}_l(\mathbf{0}, \mathbf{I})$	5	• L8: evaluate offspring
$\tilde{\mathbf{d}}_l := \mathbf{M} \tilde{\mathbf{z}}_l$	6	• L9: assemble offspring
$\tilde{\mathbf{y}}_l := \mathbf{y} + \tilde{\sigma}_l \tilde{\mathbf{d}}_l$	7	• L12f: recombine the <i>best</i> μ offspring’ $\mathbf{y}$ and σ , Eq. (2)
$\tilde{f}_l := f(\tilde{\mathbf{y}}_l)$	8	• L14: update $\mathbf{M}$ -matrix with learning rate
$\tilde{\mathbf{a}}_l := (\tilde{f}_l, \tilde{\mathbf{y}}_l, \tilde{\sigma}_l, \tilde{\mathbf{d}}_l)$	9	
End	10	
RankOffspringPopulation($\tilde{\mathbf{a}}_1, \dots, \tilde{\mathbf{a}}_\lambda$)	11	
$\mathbf{y} := \langle \tilde{\mathbf{y}} \rangle$	12	
$\sigma := \langle \tilde{\sigma} \rangle$	13	
$\mathbf{M} := \left(1 - \frac{1}{\tau_M}\right) \mathbf{M} + \frac{1}{\tau_M} \langle \tilde{\mathbf{d}} \tilde{\mathbf{d}}^T \rangle$	14	
Until (Termination_Condition)	15	

$$\tau_M := 2 + \frac{(N+1)N}{\mu} \quad (45)$$

$$\langle \tilde{\mathbf{d}} \tilde{\mathbf{d}}^T \rangle := \frac{1}{\mu} \sum_{m=1}^{\mu} \tilde{\mathbf{d}}_{m;\lambda} \tilde{\mathbf{d}}_{m;\lambda}^T \quad (46)$$

What are the Advantages of the MA-ES?

- 1 the direct MA approach is simpler than the classical CMA approach:
 - ▶ no matrix square root operation is needed, no Cholesky decomposition
 - ▶ internal algorithmic costs are reduced from $\mathcal{O}(N^3)$ to $\mathcal{O}(N^2)$ (compared to CMA-ES)
 - ▶ better suited for large-scale search space dimensionalities N
 - ▶ better suited for implementation on GPUs
- 2 reduced number of hyperparameters
- 3 greater numerical stability, regularization yet possible
- 4 simple MA-ES algorithm can be and is/was already:
 - ▶ rewritten in terms of avoiding matrix-matrix $\mathcal{O}(N^3)$ operation in the $\mathbf{M}$ update Line 14 (Slide 51) yielding the *fast* MA-version(s)
 - ▶ a starting point for *approximation schemes* to further reduce the internal costs of generating offspring (important for large-scale applications) [36]
 - ▶ a building block for developing algorithms for *constrained optimization problems* [37, 38, 39, 40, 41, 42]

$(\mu/\mu_I, \lambda)$ -fastMA-ES with Cumulative Step-Size Adaptation (CSA)

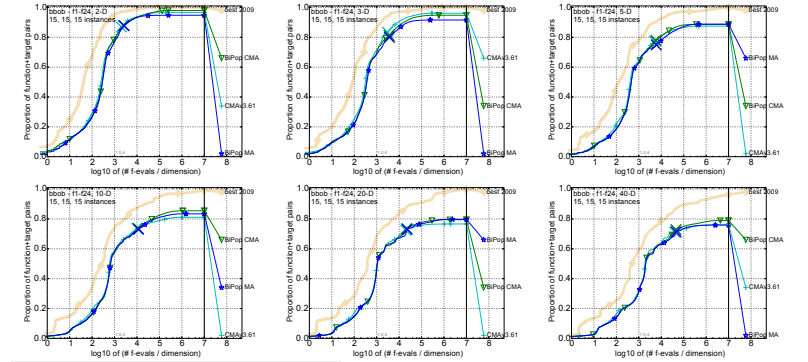
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Initialize ( $\mathbf{y}, \sigma, D, \tau_s, \tau_1, \tau_M, \mathbf{M} := \mathbf{I}, \mathbf{s} := \mathbf{1}$ ) (M1)
Repeat (M2)
  For  $l := 1$  To  $\lambda$  (M3)
     $\tilde{\mathbf{z}}_l := \mathcal{N}_l(\mathbf{0}, \mathbf{I})$  (M4)
     $\tilde{\mathbf{d}}_l := \mathbf{M}\tilde{\mathbf{z}}_l$  (M5)
     $\tilde{\mathbf{y}}_l := \mathbf{y} + \sigma\tilde{\mathbf{d}}_l$  (M6)
     $\tilde{f}_l := f(\tilde{\mathbf{y}}_l)$  (M7)
     $\tilde{\mathbf{a}}_l := (\tilde{f}_l, \tilde{\mathbf{y}}_l, \tilde{\mathbf{z}}_l, \tilde{\mathbf{d}}_l)$  (M8)
  End (M9)
RankOffspringPopulation( $\tilde{\mathbf{a}}_1, \dots, \tilde{\mathbf{a}}_\lambda$ ) (M10)
 $\mathbf{y} := \langle \tilde{\mathbf{y}} \rangle$  (M11)
 $\mathbf{s} := \left(1 - \frac{1}{\tau_s}\right) \mathbf{s} + \sqrt{\frac{\mu}{\tau_s} \left(2 - \frac{1}{\tau_s}\right)} \langle \tilde{\mathbf{z}} \rangle$  (M12)
 $\mathbf{M} := \left(1 - \frac{1}{\tau_1} - \frac{1}{\tau_M}\right) \mathbf{M} + \frac{1}{\tau_1} \langle \mathbf{M}\mathbf{s} \rangle \mathbf{s}^\top + \frac{1}{\tau_M} \langle \tilde{\mathbf{d}}\tilde{\mathbf{z}}^\top \rangle$  (M13)
 $\sigma := \sigma \exp \left[ \frac{1}{2D} \left( \frac{\|\mathbf{s}\|^2}{N} - 1 \right) \right]$  (M14)
Until(Termination_Condition) (M15)
Return( $\mathbf{y}$ ) (M16)

```

- this MA-ES represents the essence of the CMA-ES, but has internal costs of $\mathcal{O}(N^2)$
- M6: unlike σ -self-adaptation, CSA uses only one parental σ
- M12: CSA smoothed s-path cumulation
- M13: additional rank-1 $\mathbf{M}$ update
- M14: σ update
 - $D := \sqrt{N}$
 - $\tau_1 := 2N^2$
 - $\tau_s := D$
 - $\tau_M := 2 + \frac{(N+1)N}{\mu}$

- this fastMA-ES can be regarded as an *approximation* of the CMA-ES
- however, the performance differences on standard test beds including the **COCO BBOB** are not really significant (using weighted recombination)
- no big differences even for population sizes such as $\lambda = 4N^2$
- **Example:** COCO BBOB performance in BiPop-ES¹⁶ setting:



¹⁶ Matlab/Octave code can be downloaded from <https://homepages.fhv.at/hgb/downloads.html>

... for large search space dimensionalities N :

The Limited Memory MA-ES - the LM-MA-ES¹⁷

- algorithm complexity of the fast MA-ES is still of $\mathcal{O}(N^2)$ due to (M5) and (M13)
- if one wants to reduce the complexity further, one needs to *approximate* the matrix-vector and vector-vector operations in (M5) and (M13)
- an approach taking ideas from *Limited Memory BFGS* comes into mind, however, **an alternative approach will be presented here:**
- in order to approximate the $\mathbf{M}$ matrix, γ evolution path vectors $\mathbf{p}_k$ ($k = 1, \dots, \gamma$) are used
- the γ evolution paths are cumulated at different time scales
- these $\mathbf{p}_k$ vectors are used to approximate the search direction $\tilde{\mathbf{d}}$ by a weighted projection process respecting the different time scales

¹⁷I. Loshchilov, T. Glasmachers, and H.-G. Beyer [36]: Large Scale Black-box Optimization by Limited-Memory Matrix Adaptation. *IEEE Transactions on Evolutionary Computation*, 2018. DOI: 10.1109/TEVC.2018.2855049.

 $(\mu/\mu_w, \lambda)$ -LM-MA-ES¹⁸

```

Initialize ( $\mathbf{y}^{(0)}, \sigma^{(0)}, g := 0, \mathbf{s}^{(0)} := \mathbf{1}, \mathbf{p}_1^{(0)}, \dots, \gamma := \mathbf{0}$ ) (L1)
Repeat (L2)
  For  $l := 1$  To  $\lambda$  (L3)
     $\tilde{\mathbf{z}}_l^{(g)} := \mathcal{N}_l(\mathbf{0}, \mathbf{I})$  (L4)
     $\tilde{\mathbf{d}}_l^{(g)} := \tilde{\mathbf{z}}_l^{(g)}$  (L5)
    For  $k := 1$  To  $\min(g, \gamma)$  (L6)
       $\tilde{\mathbf{d}}_l^{(g)} := (1 - c_{d,k})\tilde{\mathbf{d}}_l^{(g)} + c_{d,k}(\mathbf{p}_k^{(g)\top} \tilde{\mathbf{d}}_l^{(g)})\mathbf{p}_k^{(g)}$  (L7)
    End (L8)
     $\tilde{\mathbf{y}}_l^{(g)} := \mathbf{y}^{(g)} + \sigma^{(g)}\tilde{\mathbf{d}}_l^{(g)}$  (L9)
     $\tilde{f}_l^{(g)} := f(\tilde{\mathbf{y}}_l^{(g)})$  (L10)
     $\tilde{\mathbf{a}}_l^{(g)} := (\tilde{f}_l^{(g)}, \tilde{\mathbf{y}}_l^{(g)}, \tilde{\mathbf{z}}_l^{(g)})$  (L11)
  End (L12)
RankOffspringPopulation( $\tilde{\mathbf{a}}_1^{(g)}, \dots, \tilde{\mathbf{a}}_\lambda^{(g)}$ ) (L13)
 $\mathbf{y}^{(g+1)} := \langle \tilde{\mathbf{y}}^{(g)} \rangle_w$  (L14)
 $\mathbf{s}^{(g+1)} := (1 - c_s)\mathbf{s}^{(g)} + \sqrt{\mu_{\text{eff}}c_s(2 - c_s)} \langle \tilde{\mathbf{z}}^{(g)} \rangle_w$  (L15)
For  $k := 1$  To  $\gamma$  (L16)
   $\mathbf{p}_k^{(g+1)} := (1 - c_{p,k})\mathbf{p}_k^{(g)} + \sqrt{\mu_{\text{eff}}c_{p,k}(2 - c_{p,k})} \langle \tilde{\mathbf{z}}^{(g)} \rangle_w$  (L17)
End (L18)
 $\sigma^{(g+1)} := \sigma^{(g)} \exp \left[ \frac{c_s}{2} \left( \frac{\|\mathbf{s}^{(g+1)}\|^2}{N} - 1 \right) \right]$  (L19)
 $g := g + 1$  (L20)
Until(termination condition(s) fulfilled) (L21)

```

heuristically chosen strategy parameters (for $N > 50$):

- number of evolution paths
 $\gamma := 4 + \lfloor 3 \ln N \rfloor$
- weighting constants
 $c_{d,k} := \frac{1}{1.5^{k-1}N}$
- σ -evolution path cumulation constant
 $c_s := \frac{2\lambda}{N} \quad (< \frac{1}{2})$
- $\mathbf{p}$ -evolution path cumulation constants
 $c_{p,k} := \frac{\lambda}{4^{k-1}N}$

using *weighted recombination*: $\langle \cdot \rangle_w$ and μ_{eff} , see Slide 58

¹⁸ Matlab/Octave code can be downloaded from <https://homepages.fhv.at/hgb/downloads.html>

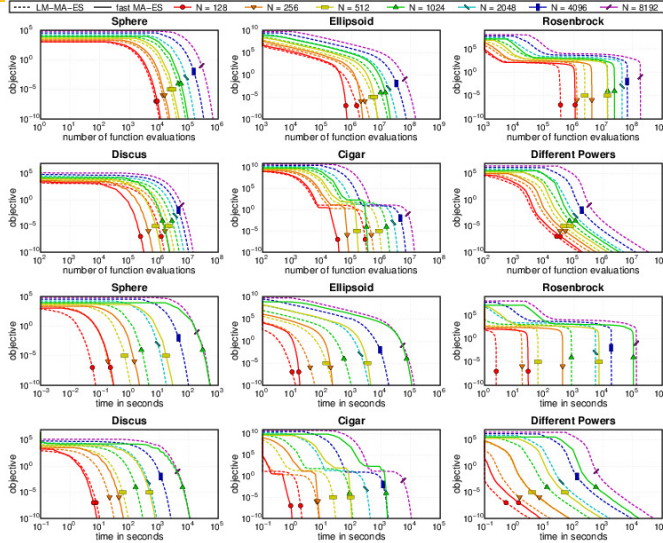


Figure 20: Performance of LM-MA-ES vs. fastMA-ES w.r.t. #-fevals and time.

Remarks (LM-MA-ES Performance)

- algorithm complexity of LM-MA-ES: $\mathcal{O}(\gamma N) = \mathcal{O}(N \ln N)$
- note, there are problem instances where LM-MA-ES outperforms MA-ES w.r.t. function evaluations (e.g. Rosenbrock $N = 256, 512$)

Remarks (additional details LM-MA-ES)

- *weighted* recombination: $\langle \tilde{\mathbf{a}}^{(g)} \rangle_w := \sum_{m=1}^{\mu} w_m \mathbf{a}_{m;\lambda}^{(g)}$
- “effective” parent population value $\mu_{\text{eff}} = (\sum_{m=1}^{\mu} w_m^2)^{-1}$
- weights w_m must fulfill $\sum_{m=1}^{\mu} w_m \stackrel{!}{=} 1$ and should not emphasize bad individuals, e.g.

$$w_m := \begin{cases} \frac{\ln(\frac{\lambda+1}{2}) - \ln m}{\sum_{k=1}^{\mu} (\ln(\frac{\lambda+1}{2}) - \ln k)}, & \text{for } 1 \leq m \leq \mu, \\ 0, & \text{otherwise} \end{cases}$$
- population sizing: $\lambda = 4 + \lfloor 3 \ln N \rfloor$ and $\mu = \lfloor \frac{\lambda}{2} \rfloor$

These recommendations have been copied from CMA-ES design [43].

Résumé

What have we learned?

- 1 ES is more than just optimization in unconstrained $\mathbb{R}^N$: $\text{CMA-ES} \subset \text{ES}$
- 2 ES design principles can also be applied to discrete problems¹⁹
- 3 especially in the theory community, “EA” is used to refer to ES in discrete search spaces, this is a *disrespect for history*
- 4 not everywhere where ES is on it is also ES in it, but the reverse is also true (there are so many “nature inspired” algorithms . . .)
- 5 state-of-the-art ES *do not trace* the natural gradient, the natural gradient approach may only serve as an *ex post* justification at most
- 6 algorithms based on natural gradient descent are only competitive when enhanced by local fitness transformations

¹⁹E.g., as early as 1988, M. HERDY [44] developed a $(5/2, 30)$ -ES including self-adaptation and recombination to tackle TSP (unpublished student research project, TU Berlin).

- 7 ES in $\mathbb{R}^N$ does not follow the gradient, its search path is more reminiscent of the serpentine when climbing a mountain
- 8 as has been shown for quadratic problems, the ES with *isotropic* mutations uses the Newton direction in expectation
- 9 correlated mutations in $\mathbb{R}^N$ are beneficial, however, explicit *learning the covariance matrix is a detour*
- 10 instead, a transformation matrix $\mathbf{M}$ can be learned directly leading to the fast MA-ES
- 11 due to its simplicity, MA-ES was/is starting point of ES developments for constrained optimization
- 12 due to the matrix operations, CMA-ES algorithms are not well-suited for large-scale problems ($N > 1000$):
- 13 instead of learning the $\mathbf{M}$ matrix and generating the mutations exactly, there is a way to do that approximately that led to the LM-MA-ES

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